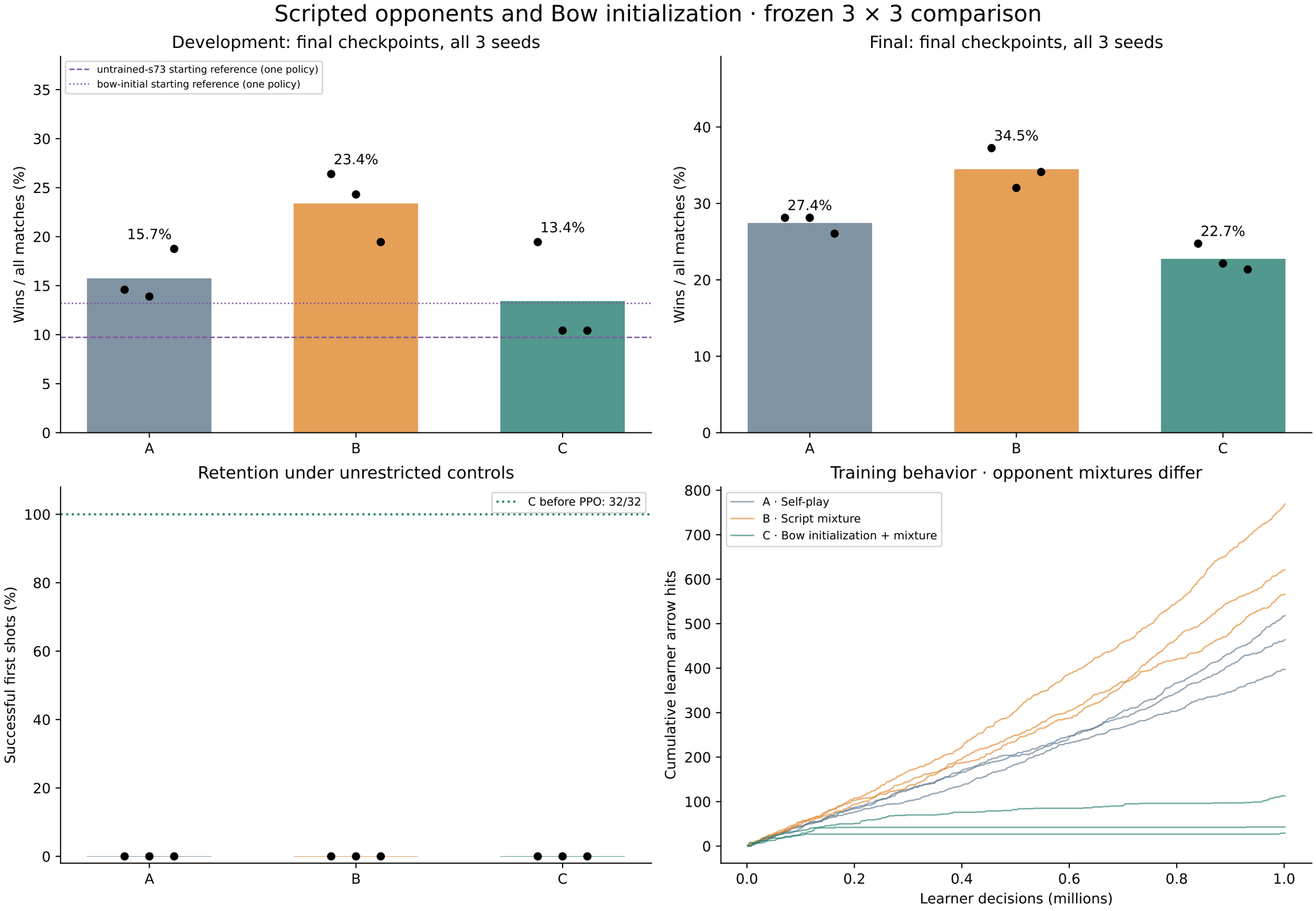
Scripted opponents and Bow initialization · frozen 3 × 3 comparison
Development: final checkpoints, all 3 seeds
untrained-s73 starting reference (one policy)
bow-initial starting reference (one policy)
Wins / all matches (%)
15.7%
23.4%
13.4%
A
B
C
Final: final checkpoints, all 3 seeds
Wins / all matches (%)
27.4%
34.5%
22.7%
A
B
C
Retention under unrestricted controls
C before PPO: 32/32
Successful first shots (%)
A
B
C
Training behavior · opponent mixtures differ
A · Self-play
B · Script mixture
C · Bow initialization + mixture
Cumulative learner arrow hits
Learner decisions (millions)